Thanh-Long V. Le

Homepage Google Scholar LinkedIn Email

Research Interests

My research centers on **large language models (LLMs)**, with an emphasis on **improving their reasoning abilities through reinforcement learning and other post-training techniques**. Recently, I have also begun exploring generative modeling, focusing on **applying reinforcement learning to diffusion and flow models**.

Education

Korea Advanced Institute of Science and Technology (KAIST AI) — **Feb. 2025 – Feb. 2027**
M.S. in Artificial Intelligence — *Seoul, South Korea*

- GPA: **4.15/4.30**
- Graduate Researcher at MLILAB

Korea Advanced Institute of Science and Technology (KAIST) — **Sep. 2019 – Feb. 2024**
B.S. in Computer Science and Artificial Intelligence — *Daejeon, South Korea*

- GPA: **3.98/4.30**
- Undergraduate Researcher at MIIL
- Graduated **Magna Cum Laude** (ranked top 8% in School of Computing)
- Minor in Business and Technology Management

Publications

No Prompt Left Behind: Exploiting Zero-Variance Prompts in LLM Reinforcement Learning via Entropy-Guided Advantage Shaping
Thanh-Long V. Le, Myeongho Jeon, Kim Vu, Viet Lai, Eunho Yang
*International Conference on Learning Representations (**ICLR**) 2026*

Design Opportunities for Explainable AI Paraphrasing Tools
Yewon Kim, **Thanh-Long V. Le**, Donghwi Kim, Mina Le, Sung-Ju Lee
*Conference on Designing Interactive Systems (**DIS**) 2025*

$(FL)^2$: Overcoming Few Labels in Federated Semi-Supervised Learning
Seungjoo Lee, **Thanh-Long V. Le**, Jaemin Shin, Sung-Ju Lee
*Conference on Neural Information Processing Systems (**NeurIPS**) 2024*

Research Experience

Adobe Research — **Aug. 2025 – Present**
Research Scientist Intern, Advised by Dr. Viet Lai — *San Jose, CA, USA*

- Led the RL-ZVP project, developing a novel algorithm that enhances LLM reinforcement learning by leveraging learning signals from zero-variance prompts.
- Currently working on improving existing RLVR techniques on text-to-image models.

Machine Learning and Intelligence Lab, KAIST AI — **Feb. 2025 – Present**
Gradute Researcher, Advised by Prof. Eunho Yang — *Seoul, South Korea*

- Conducting research on LLM post-training, including LLM reasoning and alignment through reinforcement learning/preference optimization.
- Currently focusing on augmenting RLVR with various dense learning signal to improve the reasoning ability of LLM.
- Leading a collaborative project with AITRICS on developing a large reasoning model for highly imbalanced clinical event prediction using patient EHR data.

Mobile Intelligence & Interaction Lab, KAIST — **May. 2023 – Sep. 2024**
Undergraduate Researcher, Advised by Prof. Sung-Ju Lee — *Daejeon, South Korea*

- Conducted research on natural language processing, federated learning and human-computer interaction.
- Actively participated in the Exphrase project – an AI-based paraphrasing system equipped with five supplementary features: AI Translation, AI Confidence Score, AI Explanation, Example Usages, Statistics.
- Addressed label deficiency in federated learning by implementing semi-supervised learning with adaptive thresholding and sharpness-aware regularization.

Work Experience

Luxoft – BMW Korea **Mar. 2024 – Dec. 2024**
C++ Software Engineer *Seoul, South Korea*

- Developed and optimized features for map provision application to enhance the accuracy and reliability of ADAS (Advanced Driving Assistance System) functionalities of BMW vehicles.
- Detected, analyzed, and prepared reports and fixes for vehicle anomalies by leveraging software logs and simulations.

Coc Coc Company Ltd. **Sep. 2021 – Oct. 2022**
Software Engineer *Hanoi, Vietnam*

- Maintained the functionality of three key components of Coc Coc Search's backend system – compositor, knowledge graph, and search engine.
- Enhanced the quality of Coc Coc's Answer Box by implementing multilayer rule-based question-answering system and incorporating various alternative logic to address common query patterns, resulting in a 200% increase in impression and a 30% increase in click-through rate.
- Detected and fixed a critical issue in Coc Coc Translate which related to the time-to-live setting of cached translation results, resulting in a 5-time cost reduction for external translation API calls.
- Integrated Prometheus into services to enable automatic recording of metrics, eliminating the need for manual retrieval via logs.
- Conducted a full codebase refactoring of compositor system to eliminate potential bugs and technical debt.
- Utilized SQL to collect and analyze usage data of Coc Coc Translate over the course of a year, providing insights to assist managers in decision making.

Honors and Awards

Qualcomm-KAIST Innovation Award (2023): Winner.
KAIST Dean's List (2020): Top 3% students of the department.
KAIST International Undergraduate Scholarship (2019): Full-ride scholar.
Deakin Vice-Chancellor's International Scholarship (2019): Recipient.
Vietnam National Mathematical Olympiad (2018): Silver medal.

Technical Skills

Languages: C++, C, Python, Java, Go, SQL, HTML, CSS, JavaScript.
Technologies/Frameworks: PyTorch, verl, TRL, Unsloth, Jax, Haliax, Levanter, Linux, Git, Vim, Bazel, Bash Scripting.
Certificates: 8.0 in IELTS, 1500 in SAT, 800 in SAT Math Level 2, 800 in SAT Physics.

Academic Services

Conference Reviewer: NeurIPS 2026.